

Freedom-Variant LCS — Maintenance Comparative Analysis

5 hulls | Freedom-class (Marinette/Lockheed monohull) | Comparative Maintenance & Readiness Assessment

Hulls: LCS-1 Freedom, LCS-3 Fort Worth, LCS-5 Milwaukee, LCS-7 Detroit, LCS-11 Sioux City. Data window 2012–2025.

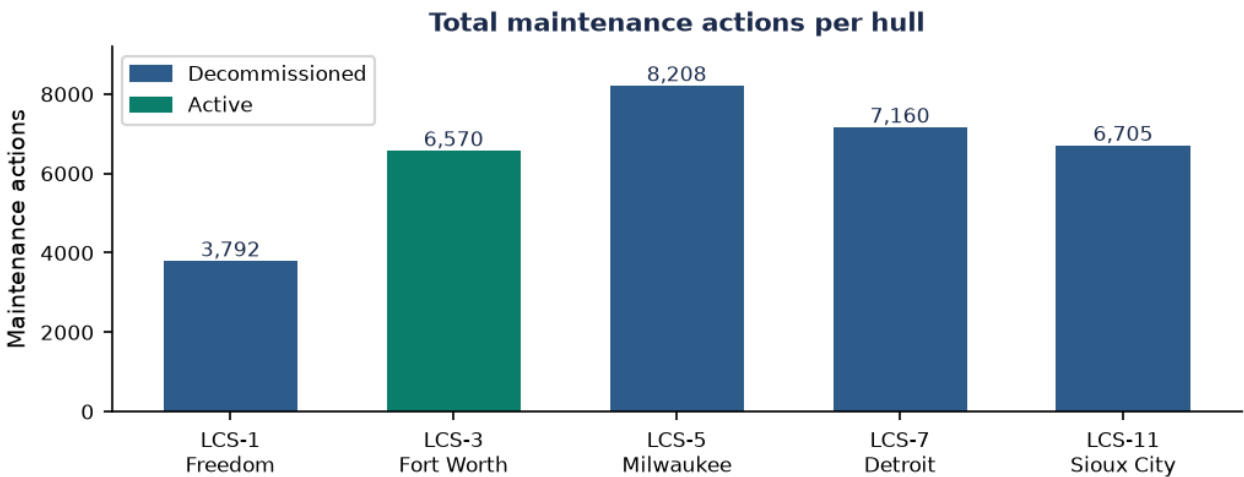
Class Overview

These five Freedom-variant hulls share an identical design — a steel semi-planing monohull with a combined diesel-and-gas-turbine plant and four ship's-service diesel generators (SSDGs). The warehouse holds **32,435 maintenance actions** across them, **\$410M** estimated cost (**\$226M** recorded actual), and **6,309 open actions** at last record. Four of the five are now decommissioned; only **Fort Worth (LCS-3)** remains active. Because the hulls are identical, the differences below reflect **operating tempo, service age, and records completeness** rather than design.

Hulls (Freedom-variant)	5	Total maintenance actions	32,435
Est. maintenance cost	\$410M	Recorded actual	\$226M
Open actions (all hulls)	6,309	SSDG generator actions	1,930
Decommissioned	4 of 5	Data window	2012–2025

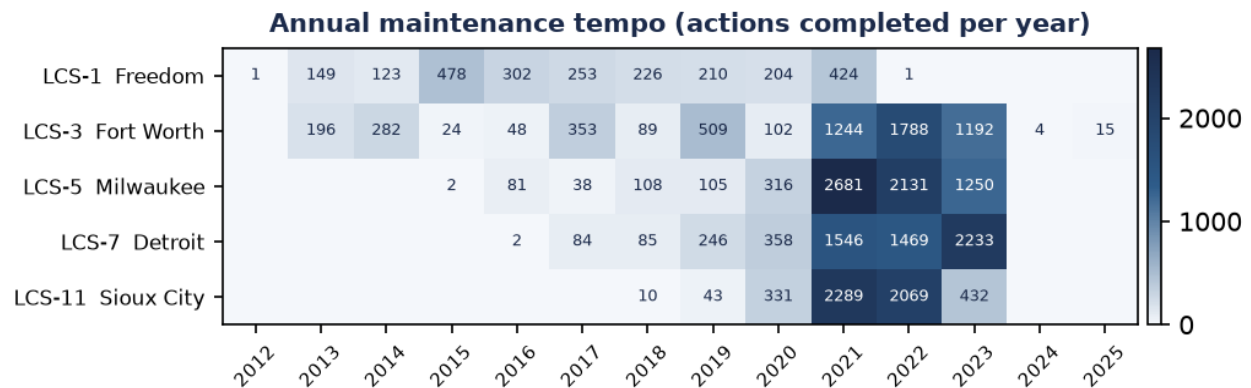
Per-Hull Comparison

Hull	Ship	Actions	% done	Open	Earliest	Latest	Est \$M	Act \$M	SSDG	Pri-1	Status
LCS-1	Freedom	3,792	63%	1,421	Jan 2012	Mar 2022	91	38	181	9	decom
LCS-3	Fort Worth	6,570	89%	724	Jan 2013	Jul 2025	88	59	299	5	activ
LCS-5	Milwaukee	8,208	82%	1,496	Nov 2015	Nov 2023	96	47	448	7	decom
LCS-7	Detroit	7,160	84%	1,137	Nov 2016	Sep 2023	61	55	442	6	decom
LCS-11	Sioux City	6,705	77%	1,531	Dec 2018	Sep 2023	74	26	560	19	decom



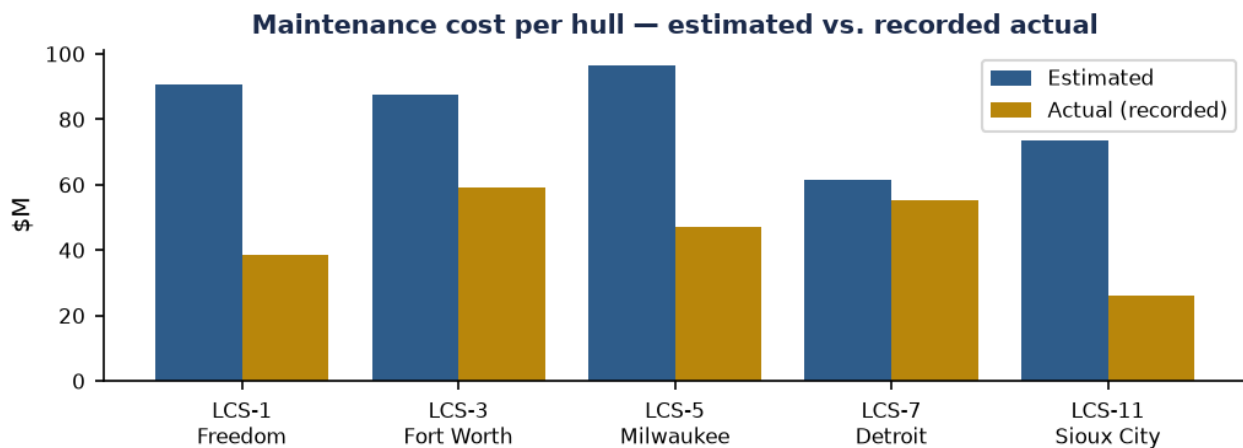
Maintenance Tempo

All five share the class life-cycle shape — quiet commissioning years, an intense 2021–2023 mid-life peak, then decline. Milwaukee (2,681 in 2021) and Sioux City (2,289) hit the sharpest peaks; Fort Worth is the outlier still generating records into 2025.



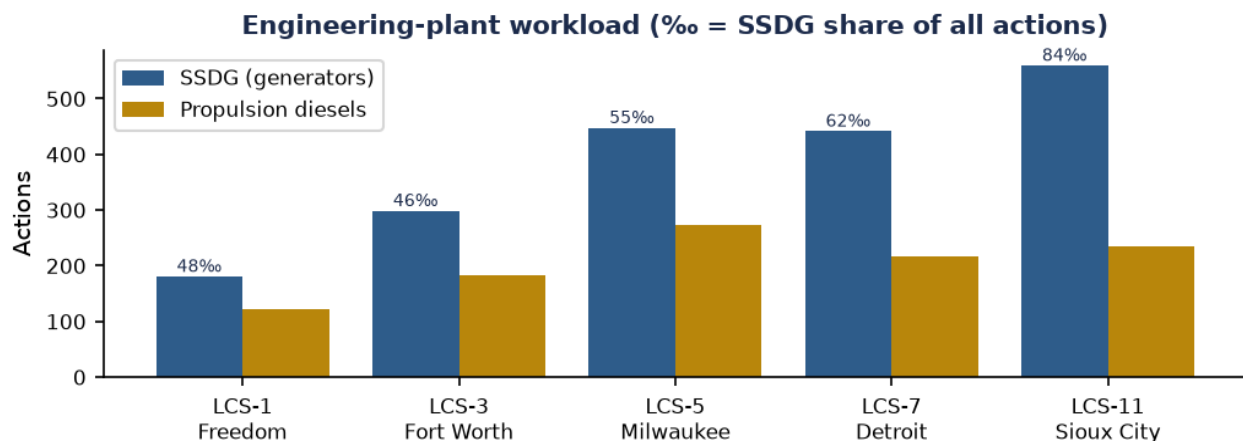
Cost & Completion

Estimated cost clusters tightly (\$61–97M); recorded actuals vary more (\$26–59M), a records-completeness artifact (uncosted open jobs) not a real cost gap — matched-pair estimate accuracy runs ~98%. Fort Worth leads completion (89%, fewest open); the decommissioned hulls froze with 1,100–1,500 open apiece.



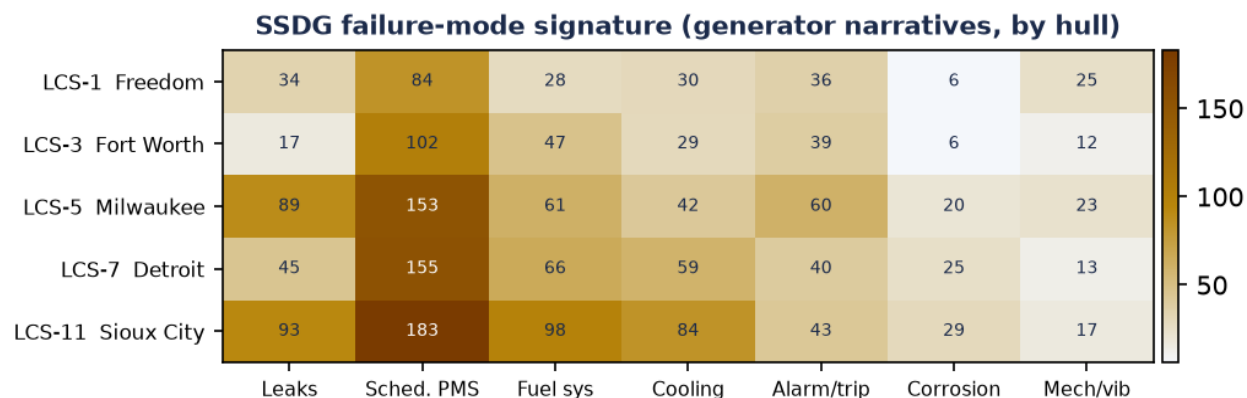
Engineering-Plant Signature — Within-Class Variation

Every hull runs the same four-SSDG plant, yet generator workload ranges **181 (Freedom) to 560 (Sioux City)** — a ~3x spread. Normalized (‰), **Sioux City spends ~84 of every 1,000 actions on generators**, class high, vs ~48 for Freedom — pointing to service age and operating tempo, not design.



SSDG Failure-Mode Signatures

Decomposing generator narratives shows the intensity gap is real wear. PMS is ~a third of SSDG work on every hull; among **unscheduled** failures, **Sioux City leads nearly every category** — leaks (93), fuel (98), cooling (84). **Fort Worth** shows few leaks (17) despite high volume — a healthier plant on the only active hull.

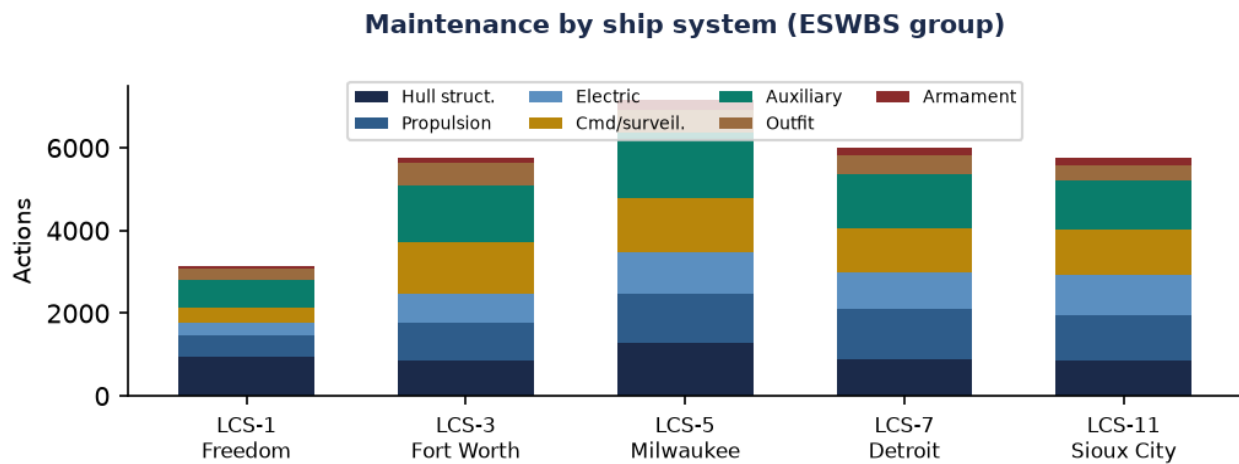
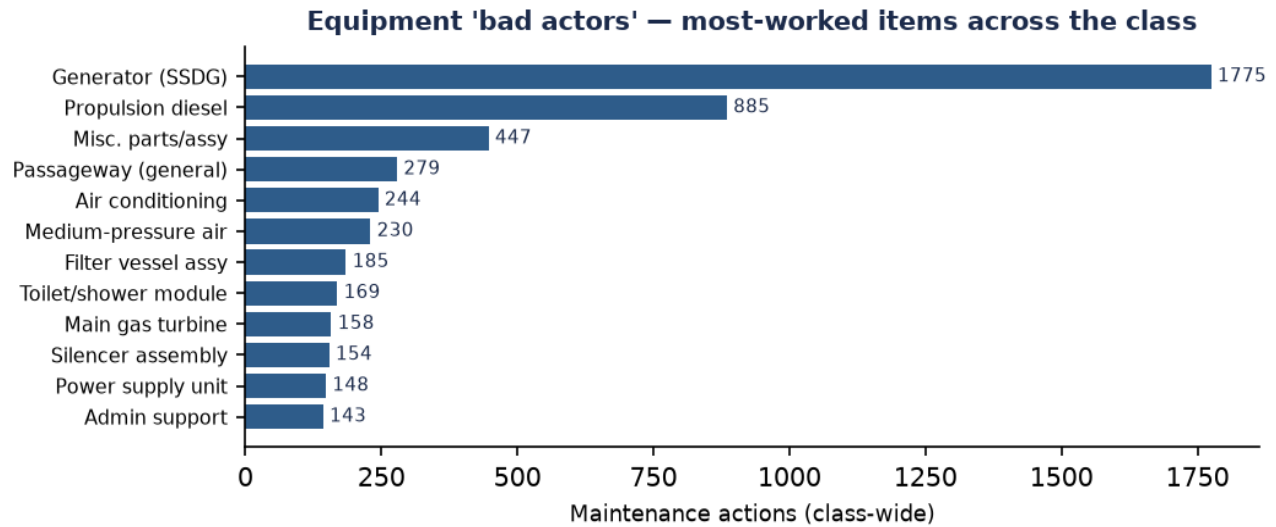


Hull	SSDG	Leaks	PMS	Fuel	Cooling	Alarm	Corros.	Mech
LCS-1	181	34	84	28	30	36	6	25
LCS-3	299	17	102	47	29	39	6	12
LCS-5	448	89	153	61	42	60	20	23
LCS-7	442	45	155	66	59	40	25	13
LCS-11	560	93	183	98	84	43	29	17

Keywords can co-occur in one narrative; categories are signatures, not a partition.

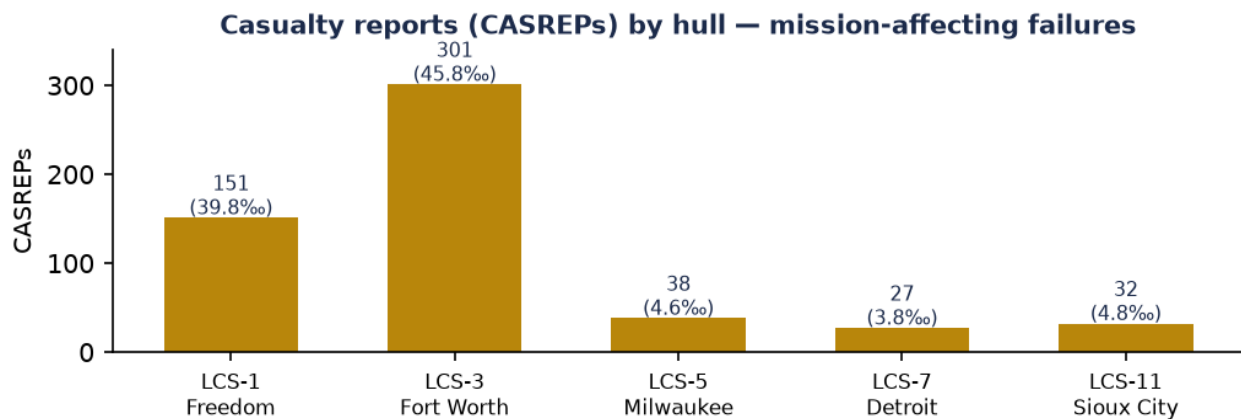
Equipment & System Reliability

Equipment bad actors. Class-wide, the engineering plant dominates: SSDG generators (1,775 actions) and propulsion diesels (885) alone outweigh the next ten equipment types combined — the 80/20 that tells a sustainment program where to focus.

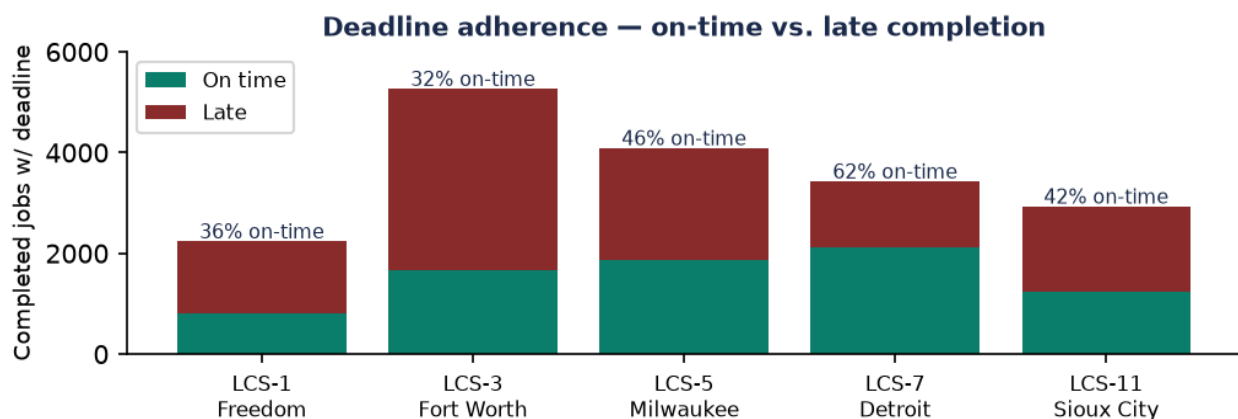


Rolled up to ship systems (ESWBS), Auxiliary and Command/Surveillance rival Propulsion for volume — the LCS's dense automation and small crew push work across every system, not just the engineering plant.

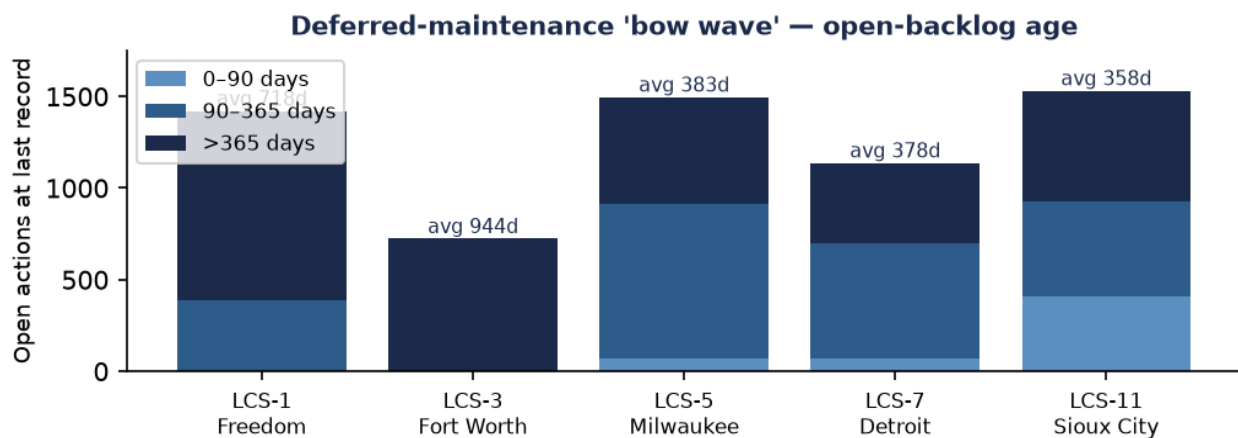
Casualty Reports & Schedule Performance



Read the CASREP chart with care. Fort Worth (45.8%) and Freedom (39.8%) show ~10x the casualty-report rate of the other three (~4–5%) — almost certainly a **change in recording practice** between earlier and later hulls, not a true reliability gap. A data-provenance flag to verify before drawing readiness conclusions.



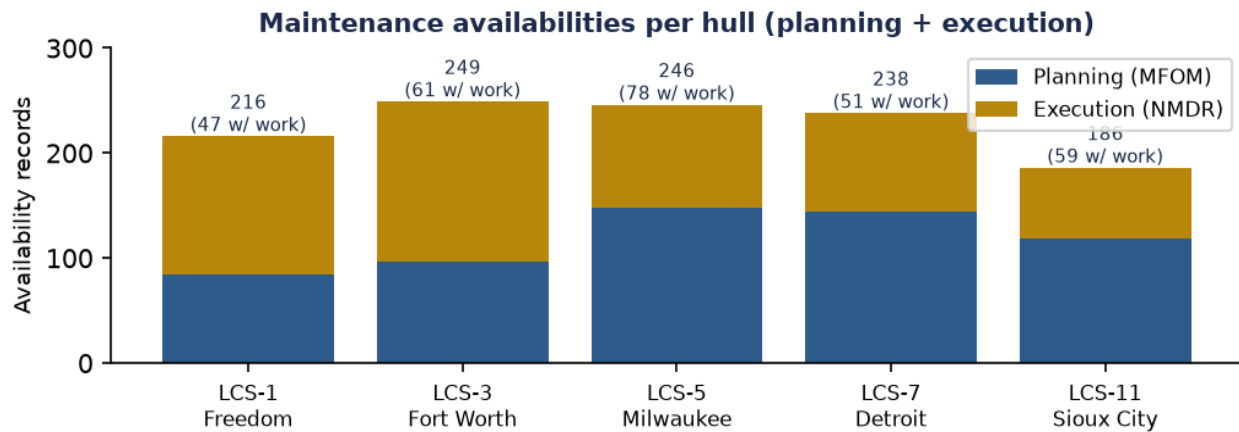
On-time completion (against the 3-M deadline) ranged 32% (Fort Worth) to 62% (Detroit); deadlines here are soft targets and routinely run over, so treat as a relative indicator.



The **deferred-maintenance 'bow wave'**: decommissioned hulls froze with 1,100–1,500 open actions, most over a year old. Fort Worth stands out — 721 of its 724 open items are >365 days old (avg 944 days): even the active hull carries a deep aged backlog behind its high completion rate.

Maintenance Availability

LCS sustainment is **availability-driven**: with a small core crew, most maintenance is executed by shore contractors during scheduled Continuous Maintenance Availabilities (CMAVs) and docking periods rather than underway. Each hull accumulated **190-250 availability records** across the planning (MFOM) and execution (NMDR) systems; of those, 47-78 per hull actually carried maintenance work, at roughly **50 maintenance actions per availability** - the batching that produces the spiky tempo seen earlier.

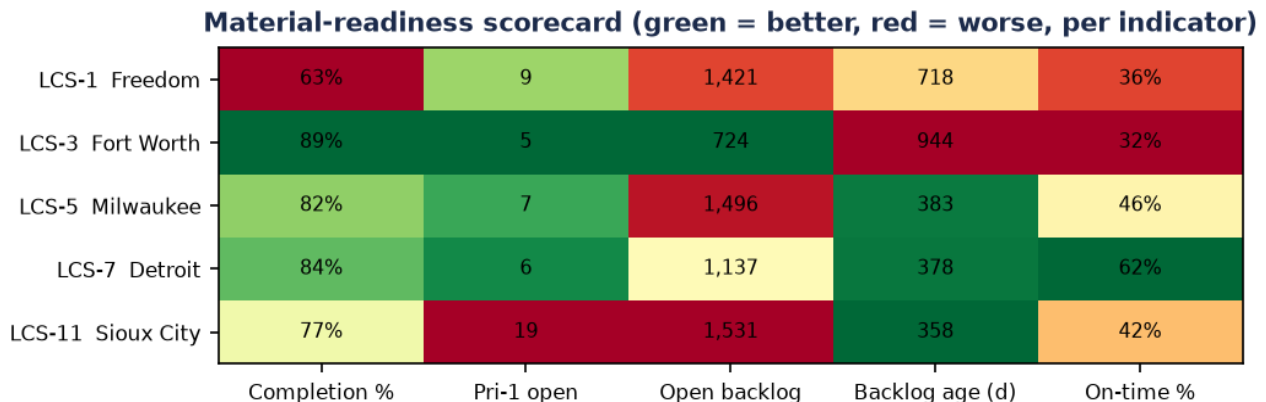


Hull	Planning (MFOM)	Execution (NMDR)	Avails w/ work	Actions / avail
LCS-1	84	132	47	58
LCS-3	97	152	61	55
LCS-5	148	98	78	49
LCS-7	144	94	51	16
LCS-11	118	68	59	46

Detroit's low actions-per-availability (16) reflects many of its actions not being linked to a specific availability record, not fewer availabilities.

Material Readiness

No single readiness rating exists in this data, but the maintenance record yields five material-readiness indicators. The scorecard normalizes each column independently (green best, red worst across the five hulls). **No hull is uniformly strong.** Fort Worth leads on completion, critical backlog and open count - yet holds the **oldest** deferred backlog (944-day average) and **worst** on-time rate. Detroit is the schedule leader (62% on-time, youngest backlog); Sioux City trails on open critical work (19) and completion. Freedom, retired earliest, shows the weakest completion (63%).



CASREP rate is excluded from the scorecard because its cross-hull comparability is doubtful (see the casualty-report note). Indicators are maintenance-derived proxies, not official readiness (C-rating) assessments.

Notable Per-Hull Findings

- **LCS-1 Freedom** — first of class, earliest and lightest record (2012–2022; 3,792 actions; lowest SSDG load at 181). Retired first.
- **LCS-3 Fort Worth** — healthiest and only active hull: 89% complete, fewest open (724), records to Jul 2025; 252 priority-1 items nearly all closed (5 open). But a deeply aged open backlog (avg 944 days) and the class's lowest on-time rate (32%).
- **LCS-5 Milwaukee** — heaviest burden: 8,208 actions, highest est cost (\$96.5M), largest single-year peak (2,681 in 2021).
- **LCS-7 Detroit** — latest-peaking (2,233 in its final year 2023); best deadline adherence (62% on-time); lowest est cost (\$61.4M).
- **LCS-11 Sioux City** — highest SSDG failure count & intensity (560; ~84%), lowest recorded actual cost (\$26.1M), two OEM-level propulsion casualties open at decommissioning.

Caveats & Interpretation Notes

- Scope: Freedom-variant hulls only (LCS-1/3/5/7/11).
- Reported cost is the estimated/actual total; the est-vs-actual spread is chiefly uncosted open jobs, not overruns (~98% matched-pair accuracy).
- SSDG intensity (per-mille) = generator actions per 1,000 total; failure-mode keywords can co-occur.
- CASREP counts likely reflect differing recording practice across hulls - comparative use only, and excluded from the readiness scorecard.
- Deadline adherence uses the 3-M deadline (a soft target); backlog age = last-record date minus deferral date.
- Availability counts span planning and execution records; availability-type detail was unavailable and a few placeholder dates were disregarded.
- Readiness indicators are maintenance-derived proxies, not official readiness (C-rating) assessments. Tempo measured by completion date; the earliest years of a hull may be under-recorded.